



الجامعة الإسلامية العالمية ماليزيا
INTERNATIONAL ISLAMIC UNIVERSITY MALAYSIA
يُونِيسَيْتِي إِسْلَامُ أَنْتَارَا بَغْسَا مِلْدِسِيَا
Garden of Knowledge and Virtue

ASSESSMENT 2 REPORT

MCTA 4363
DEEP LEARNING

GROUP: PENUNGGANG KAGGLE

TRACK 1: KAGGLE COMPETITION

Mars Environmental Dynamics Analyzer (MEDA)

**Reconstructing Missing Atmospheric Pressure Measurements from
Provenance-Preserved Rover Sensor Data**

GROUP MEMBERS	MATRIC NUMBER
MUHAMMAD AMMAR ZUHAIR BIN NOR AZMAN SHAH	2110259
MUHAMMAD BASIL BIN ABDUL HAKIM	2215315
MUHAMMAD SYAFIQ BIN NOR AZMAN	2213187

Deliverables

- Iterative model development log (experiments, hyperparameters, scores).
- Current leaderboard standing screenshot with date/time stamp.
- Documentation of techniques tried and insights gained.

Overview

The Mars Environmental Dynamics Analyzer or MEDA, records data across multiple sensors. However, due to varying polling rates and hardware constraints, the dataset contains a high frequency of missing values. Our objective was to build a machine learning pipeline that uses the available auxiliary sensor data, like solar angles and ground temperatures, to accurately predict and reconstruct the missing atmospheric pressure readings

Iterative Model Development Log

Experiment 1: Numeric Baseline (Phase 1)

- **Objective:** Establish a rapid, end-to-end pipeline and baseline Mean Squared Error (MSE) score, utilizing only readily available numerical data.
- **Feature Set:** Raw numeric columns only (e.g., `SOLAR_LONGITUDE_ANGLE`, `SOLAR_ZENITHAL_ANGLE`, `ROVER_PITCH`, `HUMIDITY_LOCAL_TEMP`).
- **Excluded Features:** Target (`PRESSURE`), Identifier (`SCLK`), and all string-based columns including temporal data (`LMST`, `LTST`).
- **Missing Data Strategy:** No manual imputation. Relied on XGBoost's sparsity-aware split finding to natively handle highly sparse auxiliary sensors (e.g., TIRS irradiance).
- **Hyperparameters:**
 - Algorithm: `XGBRegressor`
 - `n_estimators`: 100
 - `learning_rate`: 0.1
 - `max_depth`: 5
 - `eval_metric`: 'rmse'
 - `device`: 'cuda' (GPU Accelerated)
- **Result (MSE):** 8803.98
- **Notes:** Successfully established the pipeline and achieved 3rd place on the initial leaderboard. The model proved capable of predicting realistic Martian atmospheric pressures (~760 Pascals) but lacked temporal awareness.

Experiment 2: Cyclic Temporal Integration (Phase 2)

- **Objective:** Reduce MSE by equipping the model with an understanding of the Martian diurnal cycle (thermal tides).
- **Feature Set:** Baseline numeric features + Engineered Cyclic Time features (`Time_Sin`, `Time_Cos`).
- **Data Engineering:** Parsed the `LMST` (Local Mean Solar Time) string format (HH:MM:SS) into a continuous decimal hour, then applied sine and cosine transformations to map the time onto a 24-hour mathematical circle.
- **Hyperparameters:**
 - Algorithm: `XGBRegressor`
 - `n_estimators`: 200 (*Increased to allow deeper learning on new features*)

- `learning_rate`: 0.1
- `max_depth`: 6 (*Increased to capture complex time/temperature/angle interactions*)
- `eval_metric`: 'rmse'
- `device`: 'cuda'
- **Result (MSE):** 8791.79
- **Notes:** The cyclic encoding successfully eliminated the mathematical boundary at midnight (23:59 to 0:00), allowing the decision trees to model continuous atmospheric heating and cooling cycles.

Techniques Tried and Insights Gained

Technique 1: Letting the AI Handle Missing Data (Instead of Guessing)

- **What We Did (Approach):** When we looked at our data, we realized that several auxiliary sensors (like the Thermal Infrared cameras) were missing up to 90% of their readings because the sensors were either turned off or polling at different speeds. Usually, data scientists handle this by either deleting those columns or trying to "guess" the missing numbers by filling them in with averages. We chose not to guess. Instead, we fed the raw data, blanks and all, straight into our XGBoost algorithm.
- **What We Learned (Insight):** The XGBoost algorithm is naturally designed to handle blank spaces. It figures out the best way to route missing data on its own during training. If we had tried to manually fill in the blanks for sensors that were offline 90% of the time, we would have effectively been feeding the model "fake" weather data. Leaving the blanks alone gave us much more accurate, unbiased results.

Technique 2: Making Time a Circle (Cyclic Encoding)

- **What We Did (Approach):** Atmospheric pressure on Mars is heavily driven by the time of day and the sun's heat. However, if you feed time into a computer as a standard decimal number (for example, 23.99 for 11:59 PM and 0.0 for midnight), the computer thinks those two numbers are mathematically as far apart as possible. In reality, they are only one minute apart. To fix this, we used basic geometry (sine and cosine functions) to translate the timeline into a continuous 24-hour circle.
- **What We Learned (Insight):** By mapping time onto a circle, the model finally understood that midnight connects smoothly back to the start of the next day. This was a crucial fix. It stopped the model from predicting massive, impossible weather disruptions right at the stroke of midnight, allowing the math to align perfectly with actual planetary physics.

Technique 3: Using the Graphics Card for Extreme Speed (GPU Acceleration)

- **What We Did (Approach):** We specifically configured our coding pipeline to train the model using the computer's Graphics Processing Unit (GPU) instead of the standard main processor (CPU).
- **What We Learned (Insight):** Our dataset was massive, unpacking to over 1.2 gigabytes of raw telemetry data. If we had used a standard CPU, the computer would have choked on the heavy math, taking several minutes just to train the model one time. By offloading all those calculations to the graphics card, we slashed our training time down from minutes to mere seconds. This incredible speed allowed us to rapidly test out new ideas, tweak our settings, and instantly see if our changes improved the model.

Current Leaderboard

Mars Environmental Dynamics Analyzer (MEDA)

[Submit Prediction](#)

Overview
Data
Code
Models
Discussion
Leaderboard
Rules
Team
Submissions

This leaderboard is calculated with approximately 50% of the test data. The final results will be based on the other 50%, so the final standings may be different.

#	Team	Members	Score	Entries	Last	Join
1	Himanshu Dhiman		8313.049752723509	1	4mo	
2	Mubeena Hussain		8759.963132407929	1	2mo	
3	Penunggang Kaggie		8791.793925923030	2	10h	
Your Best Entry! Your most recent submission scored 8791.793925923030, which is an improvement over your previous score of 8803.984238502480. Great job! Tweet this						
	DL_Pressure_map_SAMPLE.csv		31830.593329302501	1	4mo	
4	sponish_flea		31830.593329302501	1	3mo	
	sample_submission.csv		3799158591104.9301...	1	4mo	